LOCAL QUASI-FINITENESS IS A LOCAL PROPERTY OF RING MAPS AND IS CHARACTERIZED BY IT ON AFFINE CHARTS

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 3838-3859 and proof span 3861-3894 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement and proof contain several counts of conditions by letters (a), (b), (c) referring to the cited local-property definition; these letters are the author's and are kept verbatim.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The definition of a local property of ring maps, with its three conditions (a), (b), (c) on localizations and on covers of the ring. Cited as a same-chapter definition at proof lines 4079-4081 and used throughout the first paragraph. It is the abstract notion whose conditions the author verifies for quasi-finiteness, and which then lets the same-chapter lemma on local properties be applied.
- The same-chapter statement that a local property of ring maps gives the corresponding local property of morphisms, applied with P = quasi-finite. Cited at proof lines 4102-4104. Reduces the four equivalences of the statement to the single algebraic property.
- The same-chapter characterization of quasi-finiteness at a point by the corresponding property of the local ring map. Cited at proof lines 4107-4110. Identifies the morphism-level property with the ring-level property P, completing the reduction.
- That being of finite type is already known to be a local property of ring maps, together with the fact that quasi-finiteness at a prime depends only on the local ring. Cited at proof lines 4082-4090. Supplies conditions (a) and (b) for quasi-finiteness and the local-ring reduction used in condition (c).

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 01SR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2348–2348. Same-chapter reference used in the retained author text at source line(s) 2471, 2480, 2483, 2490, 2699, 3867, 3875, 4122, 5812, 7074, 7695, 8116. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01ST: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2350–2350. Same-chapter reference used in the retained author text at source line(s) 2518, 2696, 3887, 5809, 7071, 7692, 8113. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01TH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2374–2374. Same-chapter reference used in the retained author text at source line(s) 3749, 3770, 3781, 3797, 3803, 3893, 3973, 3988, 4037, 5657, 5667, 5674, 7816, 11157, 13319, 13323, 15193, 15510. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

1. *The morphism f is locally quasi-finite.*
2. *For every pair of affine opens $U \subset X$, $V \subset S$ with $f(U) \subset V$ the ring map $\mathcal{O}_S(V) \to \mathcal{O}_X(U)$ is quasi-finite.*
3. *There exists an open covering $S = \bigcup_{j \in J} V_j$ and open coverings $f^{-1}(V_j) = \bigcup_{i \in I_j} U_i$ such that each of the morphisms $U_i \to V_j$, $j \in J, i \in I_j$ is locally quasi-finite.*
4. *There exists an affine open covering $S = \bigcup_{j \in J} V_j$ and affine open coverings $f^{-1}(V_j) = \bigcup_{i \in I_j} U_i$ such that the ring map $\mathcal{O}_S(V_j) \to \mathcal{O}_X(U_i)$ is quasi-finite, for all $j \in J, i \in I_j$.*

Moreover, if f is locally quasi-finite then for any open subschemes $U \subset X$, $V \subset S$ with $f(U) \subset V$ the restriction $f|_U : U \to V$ is locally quasi-finite.

Proof. For a ring map $R \to A$ let us define $P(R \to A)$ to mean "$R \to A$ is quasi-finite" (see remark above lemma). We claim that P is a local property of ring maps. We check conditions (a), (b) and (c) of Definition 01SR. In the proof of Lemma 01T2 we have seen that (a), (b) and (c) hold for the property of being "of finite type". Note that, for a finite type ring map $R \to A$, the property $R \to A$ is quasi-finite at $\mathfrak{q}$ depends only on the local ring $A_{\mathfrak{q}}$ as an algebra over $R_{\mathfrak{p}}$ where $\mathfrak{p} = R \cap \mathfrak{q}$ (usual abuse of notation). Using these remarks (a), (b) and (c) of Definition 01SR follow immediately. For example, suppose $R \to A$ is a ring map such that all of the ring maps $R \to A_{a_i}$ are quasi-finite for $a_1, \ldots, a_n \in A$ generating the unit ideal.

We conclude that $R \to A$ is of finite type. Also, for any prime $\mathfrak{q} \subset A$ the local ring $A_{\mathfrak{q}}$ is isomorphic as an R-algebra to the local ring $(A_{a_i})_{\mathfrak{q}_i}$ for some i and some $\mathfrak{q}_i \subset A_{a_i}$. Hence we conclude that $R \to A$ is quasi-finite at $\mathfrak{q}$.

We conclude that Lemma 01ST applies with P as in the previous paragraph. Hence it suffices to prove that f is locally quasi-finite is equivalent to f is locally of type P. Since $P(R \to A)$ is "$R \to A$ is quasi-finite" which means $R \to A$ is quasi-finite at every prime of A, this follows from Lemma 01TH. $\square$